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Project Technical Report – Web Technology
TECHNICAL ASSESSMENT 2026

Project: Agri-Nexus

Connecting Rwanda's Agriculture — from soil to sale

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List of Abbreviations

- AI — Artificial Intelligence
- API — Application Programming Interface
- DB — Database
- ERD — Entity-Relationship Diagram
- ICT — Information and Communication Technology
- IoT — Internet of Things
- NAEB — National Agricultural Export Development Board
- QR (code) — Quick Response code
- RWF — Rwandan Franc
- UI — User Interface

Chapter 1. Introduction

1.1. Historical Background of the Case Study

Rwanda's economy and rural livelihoods depend heavily on agriculture. Agri-Nexus takes its name from "Agri" (Agriculture) and the Latin "Nexus" (binding together) — a platform that connects farmers, buyers, labourers, government bodies and technology partners, who today largely operate in isolation from one another.

The sector faces recurring structural problems: poor market-price transparency, inefficient resource use, weak product traceability, labour shortages, and growing climate unpredictability. Agri-Nexus is a single web-based ecosystem designed to address these problems directly.

1.2. Problem Statement

1. Information gap on market prices: Small-scale farmers often don't know current crop prices and sell at a loss to intermediaries.

2. Improper resource management: High fertiliser and energy costs make blanket-spraying entire fields unaffordable.

3. Lack of product traceability: Consumers want to know where food comes from, but there is no easy digital way to verify it.

4. Labour shortage and poor coordination: Trained workers leave the sector, and the remaining labour force is poorly coordinated.

5. Unpredictable climate conditions: Climate change has made planting calendars unreliable, with new droughts, pests and diseases.

1.3. Objectives of the Project

- A price-transparent marketplace with live, government-verified prices connecting farmers directly to buyers.
- Satellite-based field "heat maps" showing exactly where nitrogen, water or other nutrients are needed.
- QR-coded products revealing harvest date, growing method and producer information.
- A labour connection board and map-based task assignment ("Farm Operating System").
- An AI advisory service for crop/pest diagnosis and real-time weather alerts.

1.4. Expected Outcomes

- Fairer prices for farmers selling directly to buyers.
- Less fertiliser, water and energy waste through targeted field management.
- Greater consumer trust in traceable Rwandan produce.
- Better-coordinated agricultural labour.
- Farmers more resilient to climate shocks.

1.5. Project Rationale (Importance to the Community)

Most Rwandan households depend on agriculture, so gains in price transparency, resource efficiency, traceability, labour coordination and climate resilience have an outsized impact on rural incomes and food security. Agri-Nexus aims to make the agricultural value chain fairer, more efficient and more climate-resilient.

Chapter 2. System Analysis and Design

2.1. System Analysis

2.1.1. Intended Users of the Project (Internal Users)

Produce buyers and wholesalers: Verify crop quality and origin via QR traceability before buying in bulk.

Agricultural labourers: View assigned zones and conditions for the day.

Farm owners / managers: Check crop health via satellite data and assign daily tasks.

Farmers (sellers): List products, view live prices, get AI crop advisory, monitor field heat maps.

Visitors: Prospective users who register interest before creating a full account.

2.1.2. Intended System Partners (External Users)

Financial institutions and insurers: Use platform data to offer farmers better loan rates or faster payouts.

Government and meteorological agencies: Supply weather alerts and the official market-price data feed.

Satellite and imagery providers: Supply the aerial imagery used to generate field heat maps.

2.2. System Design

1. High-Level Architecture

The platform has a Web platform (this assessment's deliverable), a companion Mobile platform for field use, and a Satellite/IoT layer feeding field-condition data. All three connect through a common API layer backed by a central database, and out to external partners (government/meteorological agencies, financial institutions, satellite-imagery providers). Each layer pairs with an AI service for advisory, forecasting or field analysis.

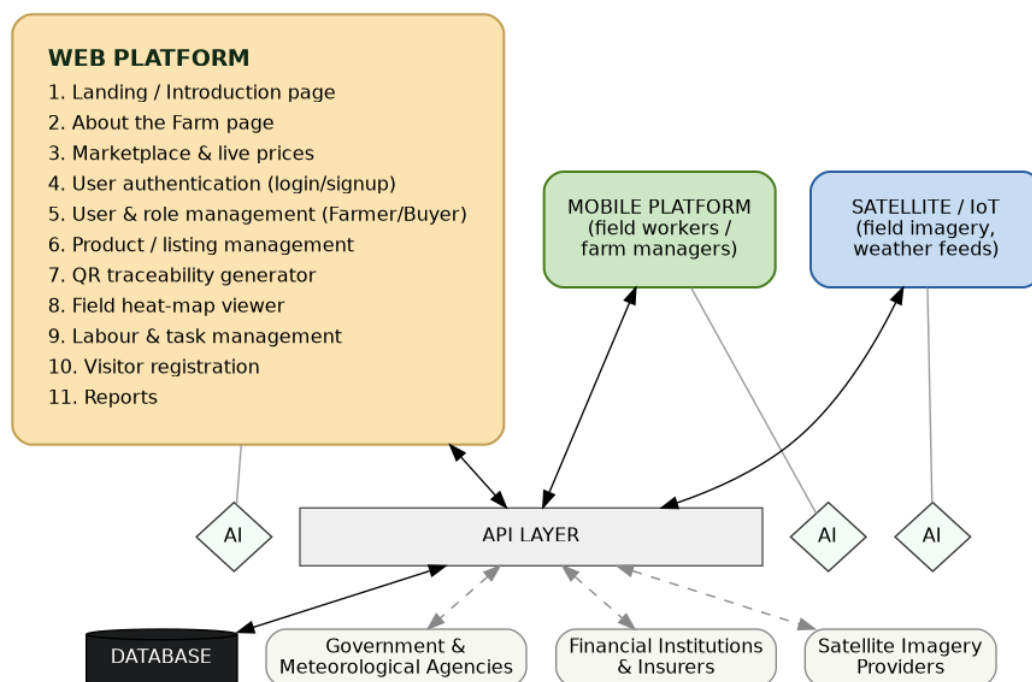


Figure 1: High-level system architecture of Agri-Nexus

2. Database Design with ERD and Relationships

A central User entity (farmer or buyer, by role) links to Product Listings, Orders, Farm Fields (heat-map data), Labour Tasks, and an AI Advisory log. A Market Price entity holds the government price feed, and a Visitor entity stores Visitor Registration submissions.

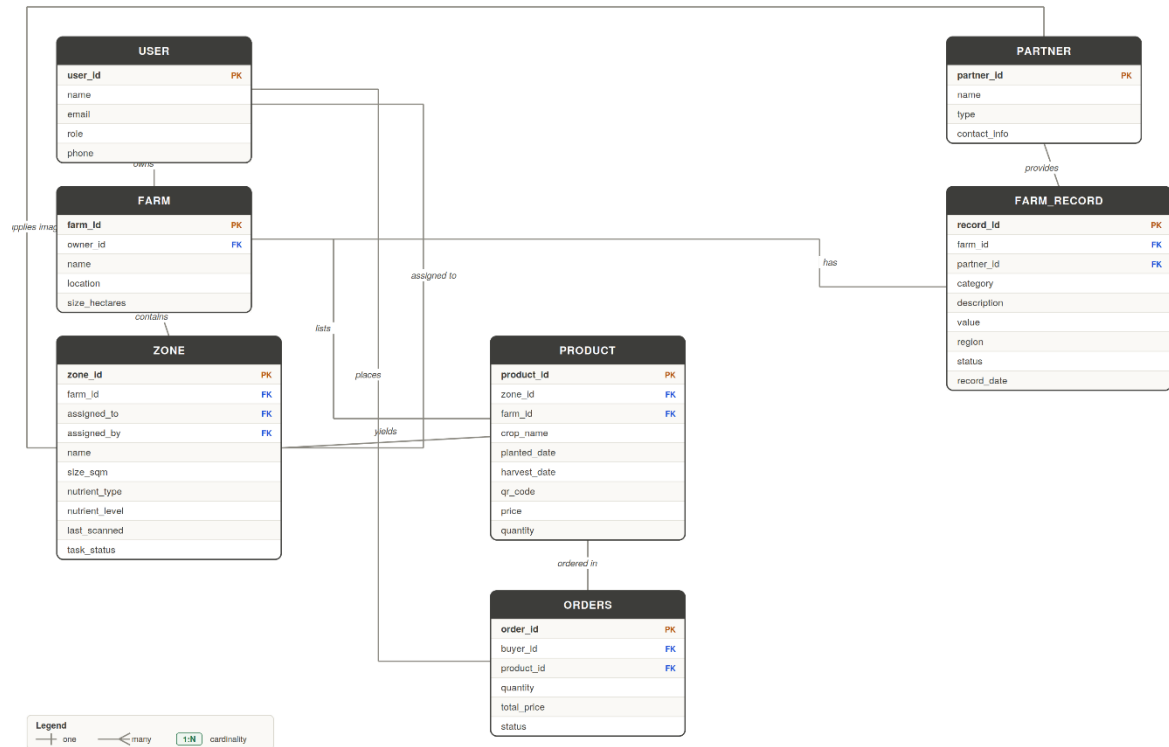


Figure 2: Entity-Relationship Diagram (ERD) of the Agri-Nexus database

Chapter 3. Implementation

3.1. Introduction

This chapter presents the pages implemented so far: Introduction/Landing, About the Farm, Agri-Nexus Market, and the Login, Create Account, Forgot Password and Visitor Registration pages. Note: the pages were rendered using only their inline styles; the external CSS and image assets were not included in this submission, so some visual polish is not reflected below.

3.2.1. Introduction / Landing Page

The entry point of Agri-Nexus: a hero section with the platform name and tagline, and a nav bar linking to Market, About Farm and Login.



Figure 3: Introduction / Landing page — hero section

A problem/solution table lays out the five core problems the platform addresses alongside the feature that solves each one.

No	PROBLEM	SOLUTION
1	we have One of the oldest problems in farming of the Information Gap . Small-scale farmers often don't know the current market price of their crops, leading them to sell to intermediaries at a loss.	This web application will provide a marketplace that tracks regional commodity prices and connects farmers directly consumers. The information about these prices will be provided by the government, and this will fix problem of price transparency
2	we have to deal with unproper Resource management . With fertilizer and energy prices are being to high(expensive), we have to get solutions dealing with this. For example, spraying an entire field the same way is no longer affordable.	This web application will integrate with satellite imagery to provide "heat maps" of a farm. A farmer logs into the website to see exactly which square meter of their land needs nitrogen or water or any other nutrient. In case of supplying all the field, farmer can supply the nutrient to that small part in need (according to the example provided above).
3	we will need digitalized market of agriculture products. Modern consumers need to know exactly where their food came from and how it was grown. These consumers may need to be clients to the provider because of his fresh production or need to know how they can grow that product being consumed.	This web application will require that each product on its market will have a QR code so that A buyer scans a QR code on product and get information about the product like harvest time and information about the producers.
4	we have seen that many people who have studied Agriculture in their university do not make it as a career, they deviate to other domains and this makes a shortage of labours. And other problem is that the ones in shortage do not work efficiently (lack of Team coordination) .	This web application will provide a connection area for the labours and Employers. And will have a "Farm Operating System" where supervisors can assign tasks using a map.so the few workers available are used effectively.
5	Climate change has made traditional planting calendars unreliable . In 2026, farmers are facing unpredictable droughts and new pest infestations. This creates some new complicated challenges like diseases attacking crops and other non-friendly conditions.	This web application will provide an AI-driven advisory (specialised on Agriculture) that you can tell the condition your farm or crops are in and that AI can advise you what to do. And this AI advisory can provide Alerts of weather forecast to the web application users.

Figure 4: Introduction / Landing page — problem & solution table

The page closes with content on Rwanda's key staple and export crops, and a call-to-action toward the Visitor Registration form.

- Coffee

Coffee is Rwanda’s leading agricultural export and a major driver of national economic stability, hitting a historic record of \$148.6 million in revenue in 2025. The industry supports over 400,000 smallholder farming households, primarily growing high-quality Arabica beans that are processed in over 300 washing stations across the country. Because the government has focused on “specialty coffee” rather than bulk volume, Rwandan beans fetch premium prices in North America, Europe, and Asia, making it a critical tool for poverty reduction and increasing the income of rural families.



Coffee from Rwanda

Coffee farm in Rwanda

- Tea

Tea is a consistent and high-performing export that generates nearly \$100 million annually, prized globally for its high quality and distinct flavor profile due to Rwanda's volcanic soils and high altitudes. Unlike many seasonal crops, tea provides a year-round income for farmers and workers in the Northern, Western, and Southern Provinces, where large estates and cooperatives dominate the landscape. Rwandan tea is a staple at the Mombasa Tea Auction, where it frequently commands the highest prices in the East African region, making it a reliable pillar of the country's trade balance.

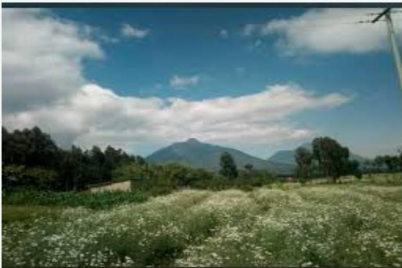


Quality tea from Rwanda

Tea farm in Rwanda

- Pyrethrum

Rwanda is the world’s second-largest producer of pyrethrum, a natural insecticide extracted from chrysanthemum flowers grown in the rich soils of the volcanic Virunga region. This niche but high-value crop accounts for roughly 15% of the global market share, providing a specialized export that serves the international organic pesticide and pharmaceutical industries. By focusing on sustainable cultivation in districts like Musanze and Nyabihu, Rwanda has turned pyrethrum into its third most important traditional cash crop, offering a vital and unique revenue stream that diversifies the agricultural sector away from beverages alone.



Pyrethrum farm in mountain valley

Farmers in a Pyrethrum farm

Figure 5: Introduction / Landing page — key & export crops section

3.2.2. About the Farm Page

Presents Agri-Nexus as a digital ecosystem for farmers, workers and buyers, breaking down its five services — Marketplace, Field intelligence, Traceability, Labour, and AI Advisory — each tagged with the user groups it serves.

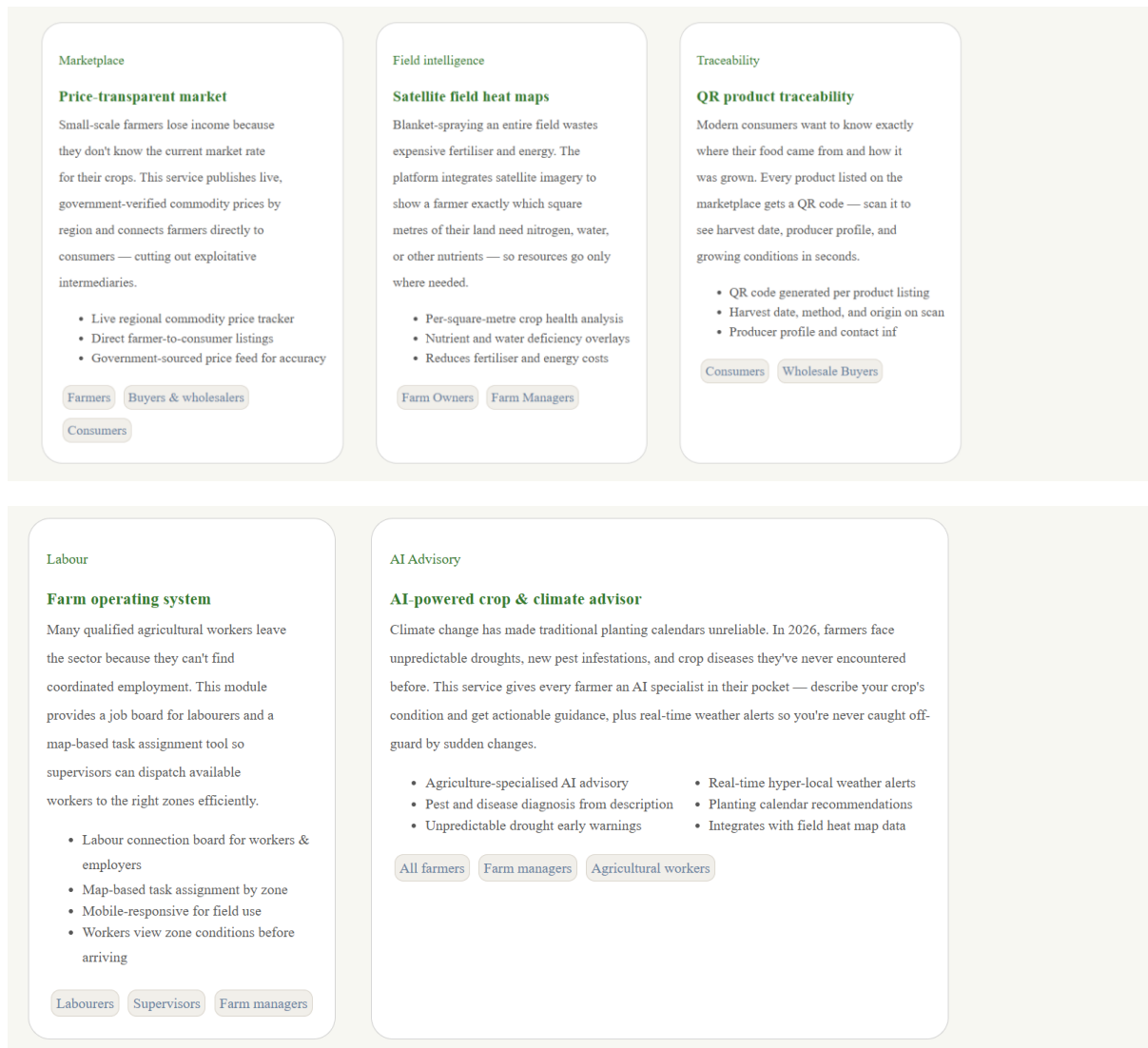


Figure 6: About the Farm page — services offered.

3.2.3. Agri-Nexus Market Page

The core marketplace experience. The hero highlights a "live" market with a running "Today's Prices" card for stable commodities-the government sourced price feed from chapter 1.

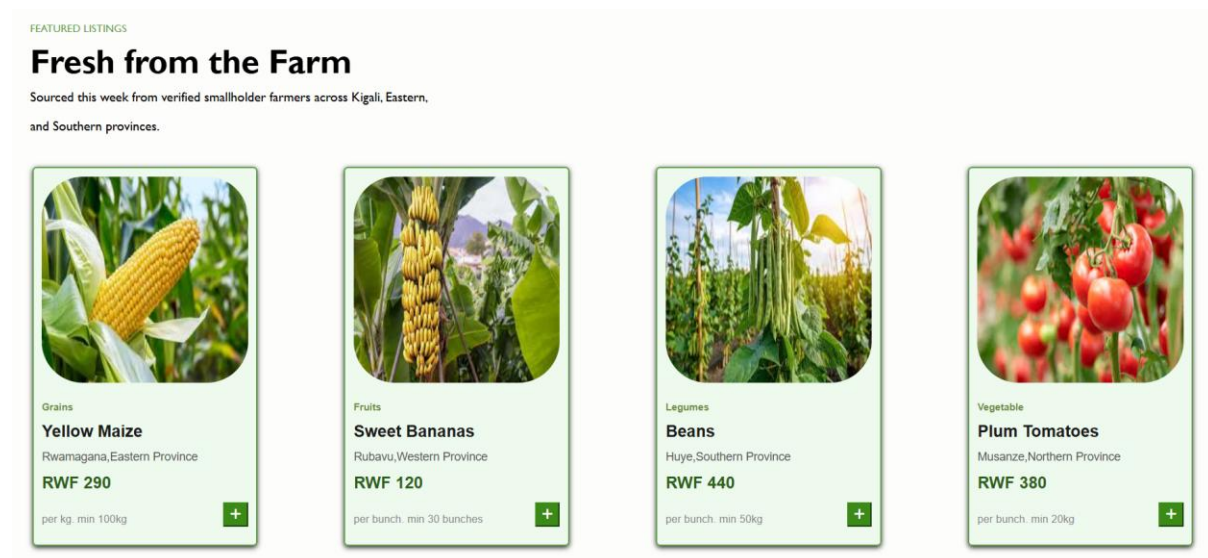


Figure 7: Agri-Nexus Market page — hero and live prices

More listings follow, plus a "Doorstep in 4 Steps" section explaining the buying process end to end.

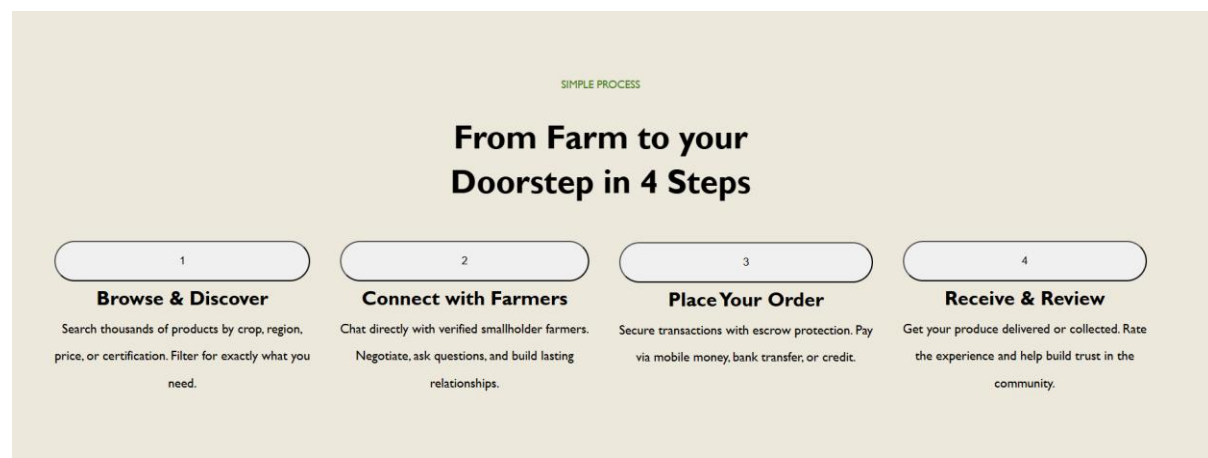


Figure 8: Agri-Nexus Market page — 4-step process

The page ends with a call to action toward account creation and a footer summarising the site's sections.

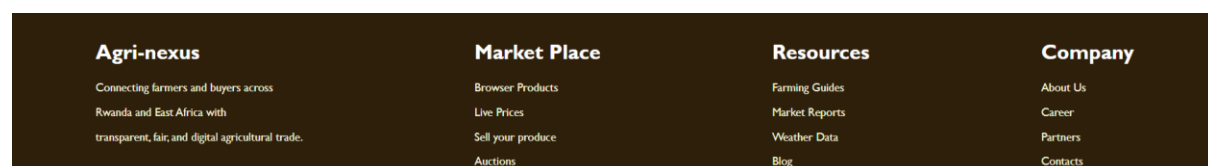


Figure 9: Agri-Nexus Market page — call to action & footer

3.2.4. Login Page

A minimal authentication form (e-mail, password) with links to Forgot Password and Create Account.

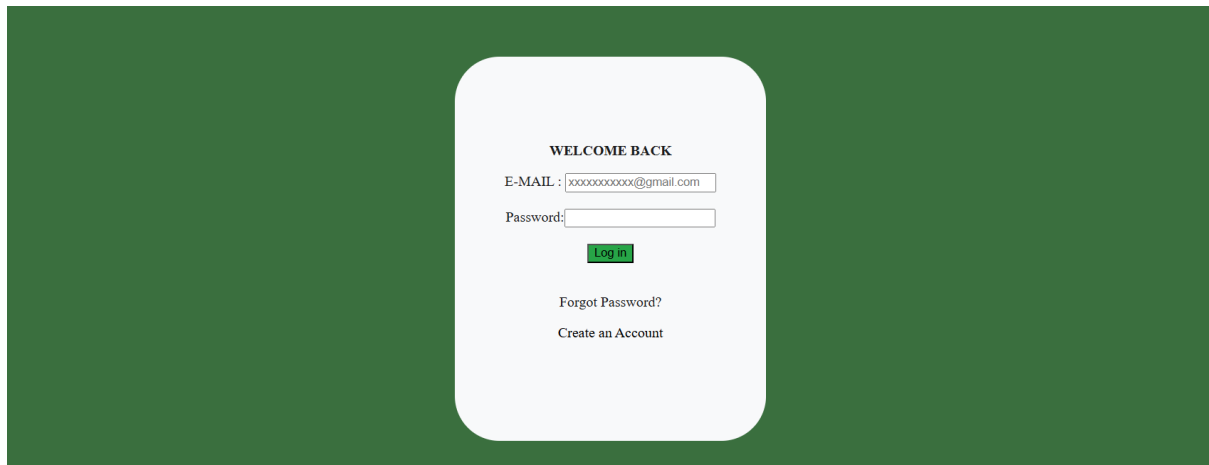
The login page features a dark green background. In the center is a white rounded rectangle containing the login form. At the top of the form is the heading "WELCOME BACK". Below it are two input fields: "E-MAIL:" with the placeholder "xxxxxxxxxx@gmail.com" and "Password:". A green "Log in" button is positioned below the password field. At the bottom of the form are two links: "Forgot Password?" and "Create an Account".

Figure 10: Login page

3.2.5. Create Account Page

Captures account type (Farmer/Seller or Buyer), full name, e-mail, phone, province and password. Client-side JavaScript checks the password is at least 6 characters before confirming.

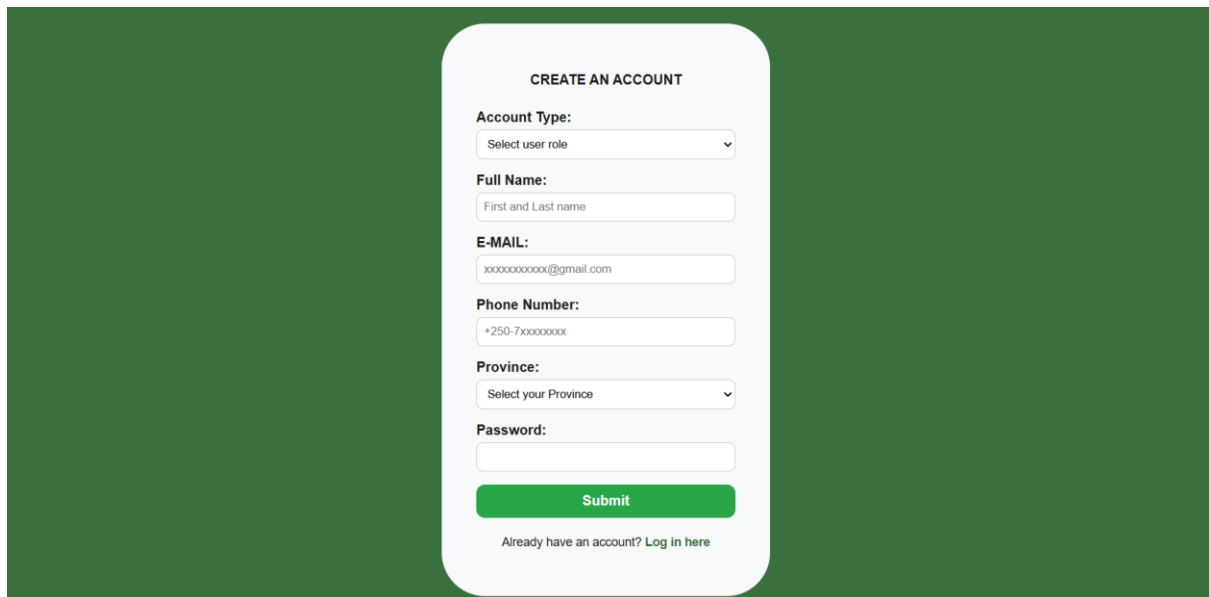
The create account page has a dark green background with a white rounded rectangle in the center for the form. The heading "CREATE AN ACCOUNT" is at the top. The form includes several fields: "Account Type:" with a dropdown menu showing "Select user role"; "Full Name:" with a text input field showing "First and Last name"; "E-MAIL:" with a text input field showing "xxxxxxxxxx@gmail.com"; "Phone Number:" with a text input field showing "+250-7xxxxxxxx"; "Province:" with a dropdown menu showing "Select your Province"; and "Password:" with a text input field. A green "Submit" button is located below the password field. At the very bottom of the form is a link: "Already have an account? Log In here".

Figure 11: Create Account page

3.2.6. Forgot Password Page

Let a user recover access with e-mail, current password and phone number, then request a reset code by SMS or e-mail.

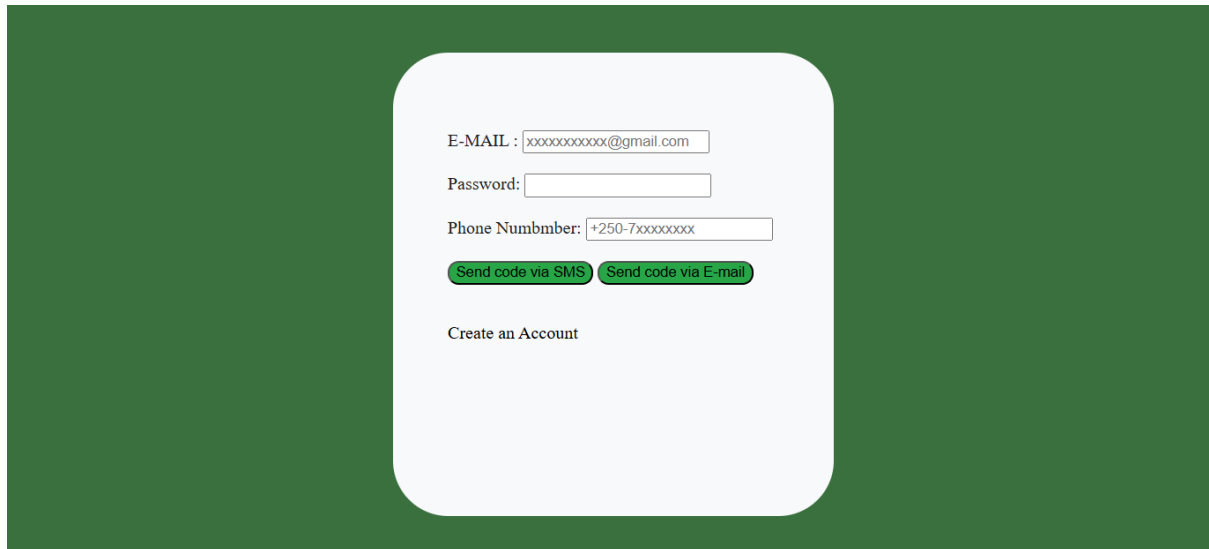
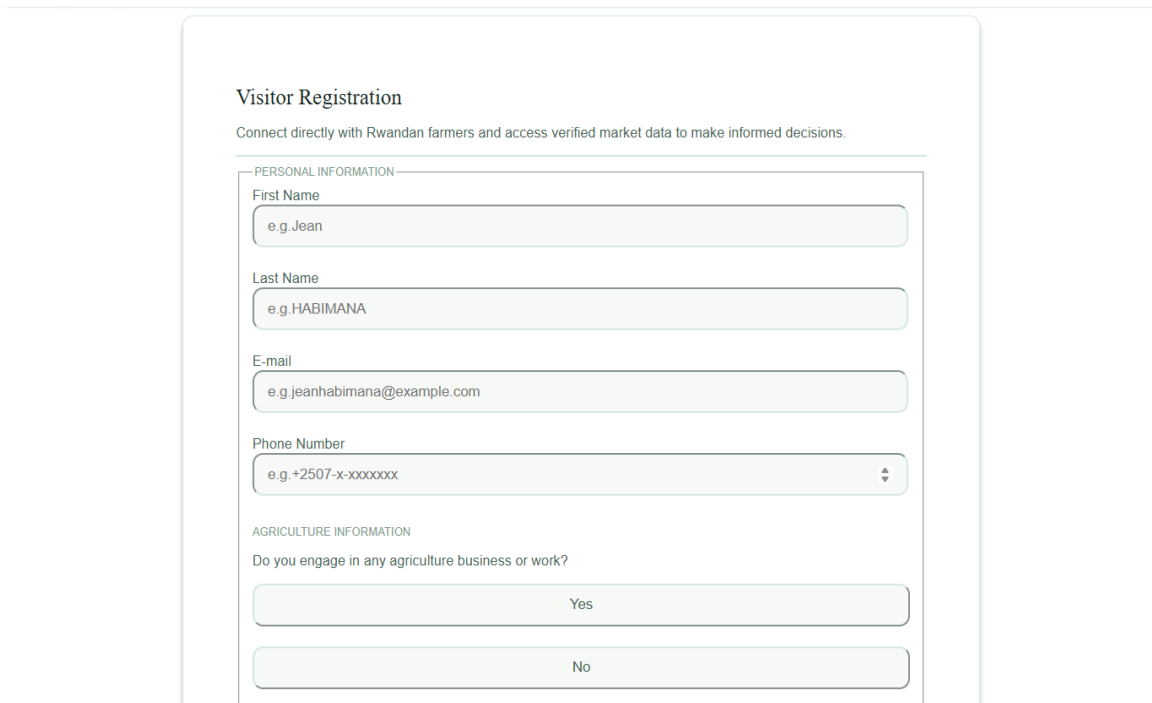
The image shows a 'Forgot Password' page with a dark green background. In the center is a white rounded rectangle containing the form. The form has three input fields: 'E-MAIL' with the placeholder 'xxxxxxxxxx@gmail.com', 'Password', and 'Phone Number' with the placeholder '+250-7xxxxxxxx'. Below these are two green buttons: 'Send code via SMS' and 'Send code via E-mail'. At the bottom of the white box is a link that says 'Create an Account'.

Figure 12: Forgot Password page

3.2.7. Visitor Registration Page

For prospective users not yet ready to create an account collects contact details, agriculture-involvement status, features of interest, notes, and a data-processing consent checkbox.

The image shows a 'Visitor Registration' form. At the top, the title 'Visitor Registration' is followed by a subtitle: 'Connect directly with Rwandan farmers and access verified market data to make informed decisions.' The form is divided into two sections. The first section, 'PERSONAL INFORMATION', contains four input fields: 'First Name' (placeholder: e.g. Jean), 'Last Name' (placeholder: e.g. HABIMANA), 'E-mail' (placeholder: e.g. jeanhabimana@example.com), and 'Phone Number' (placeholder: e.g. +2507-x-xxxxxx). The second section, 'AGRICULTURE INFORMATION', contains a question 'Do you engage in any agriculture business or work?' with two radio button options: 'Yes' and 'No'.

Great! AgriNexus is built for people like you. You will get access to market prices, satellite farm tools, and direct connections with buyers.

FEATURES OF INTEREST

- ☐ Market price tracking & transparency
- ☐ QR code crop origin & harvest info
- ☐ Satellite field health maps
- ☐ AI-driven crop advisory & weather alerts
- ☐ Labour management & farm updates

ADDITIONAL NOTES

Any specific requirements or questions?

Write your notes here...

☐ I consent to the processing of my personal data in accordance with the privacy policy.

Submit

Figure 13: Visitor Registration page

Chapter 4. Conclusion and Recommendation

The public-facing and account pages of Agri Nexus are implemented: Introduction/Landing, About the Farm, Agri-Nexus Market, Login, Create Account, Forgot Password, and Visitor Registration. These cover the platform's value proposition, live-price marketplace browsing, and the full authentication/onboarding flow. Still to be built: role-based dashboards, back-end data persistence, the satellite heat-map viewer, QR traceability generator, labour task-assignment tool, and AI advisory service — roughly 40–50% of the full scope, with the front end substantially complete.

Recommendations

To the lecturer: Guidance on integrating a live government price feed and a satellite-imagery API would help move the Marketplace and Field Intelligence modules to working, data-backed features.

To the school, on time constraints: A platform of this scope benefits from being scheduled across more than one assessment cycle or scoped to a subset of modules per cycle.

To the partners: Early engagement with cooperatives, price-data sources and imagery providers is recommended, since the price-transparency and field-intelligence features depend on these feeds.

To users and beneficiaries: Farmers and labourers will need simple onboarding materials, and consumers should be shown how to use QR traceability at the point of purchase.

Appendix.

<https://github.com/Jam906-cpu/Agri-Nexus-Project>

